EDUCATION

Carnegie Mellon University, Pittsburgh, PA
Master of Science in Computer Science, School of Computer Science *August 2026 – January 2028 (expected)*
Coursework: Computer Systems
Current Coursework: Distributed Systems, Generative AI, Algorithms
University of Maryland, College Park, MD
Bachelor of Science in Computer Science, Minor in Statistics *May 2026*
GPA: 4.0/4.0 | Summa Cum Laude
Coursework: Operating Systems, Databases, Networks, Parallel Computing, Machine Learning, Natural Language Processing

EXPERIENCE

Backend Software Engineer Intern — Creator Assistant *May 2026 – August 2026*
ByteDance (TikTok Shop)

- Integrated an internal agent platform into Creator Assistant's Go backend, providing the ReAct agent with configurable skills, MCP tools, and knowledge; built live-data MCPs and optimized tool-schema loading to reduce token usage.
- Designed and built an agent evaluation harness with automated RPC execution, tool/skill tracing, and blind response review; incorporated real creator conversations alongside curated test cases.
- Identified an agent observability bottleneck and built a trace viewer that renders model context, tool calls, and internal traces in a readable chat format; developed it into a debugging tool used across the team to diagnose race conditions, tool failures, and response latency.
- Creator Assistant launched in production for all TikTok Shop Affiliate creators.

Student Developer — Distributed Multi-Region Database *September 2025 – June 2026*
Data Systems Lab at Maryland (DSLAM), University of Maryland (Supervised by Dr. Daniel Abadi)

- Implemented strict two-phase locking with wait-for graph deadlock detection in a C++ distributed multi-region database.
- Built distributed transaction validation/commit coordination and asynchronous WAL replication across database nodes.

Backend Software Engineer Intern — Journey Builder *May 2025 – August 2025*
ByteDance (TikTok Shop)

- Built Journey Builder's React UI to configure onboarding flows for 250K+ new TikTok Shop creators each month.
- Extended Go/Kitex backend services and implemented GORM/MySQL metrics endpoints for Journey Builder.
- Deployed services through internal CI/CD pipelines and collaborated with infrastructure teams to resolve production issues.

Undergraduate Student Researcher *February 2025 – May 2025*
Department of Computer Science, University of Maryland (Advisor: Dr. Amol Deshpande)

- Designed and implemented an importance sampling estimator for aggregate queries with expensive LLM predicates.
- Benchmarked against uniform and two-stage stratified sampling in simulation, reducing estimator variance while analyzing robustness and rare high-variance outliers. Research Report

PROJECTS

UMIACS Wiki Chatbot | *Developer* | *Technical Report* *August 2024 – December 2024*

- Developed a chatbot for the University of Maryland Institute for Advanced Computer Studies (UMIACS) Wiki to automate responses to common support questions as part of a six-person team.
- Built a RAG pipeline with LangChain and Qdrant, experimenting with embedding models and chunking strategies to improve retrieval quality.
- Fine-tuned LLaMA3-8B-Instruct with LoRA on UMIACS question-answer data.

Doodles and Character Classifier | *Full-Stack Developer* | *Live Demo* *July 2024 – August 2024*

- Built a web application using React and Node.js that classifies user-drawn images in real time.
- Trained PyTorch models on Quick, Draw! and EMNIST; implemented browser-side and server-side inference.

TECHNICAL SKILLS

Languages: Python, Go, C++, JavaScript, Java, C, SQL
Frameworks & Tools: Git, Linux, Docker, AWS, Thrift, Protobuf, Kitex, GORM, Node.js, Express.js, React
AI Systems & ML: Model Context Protocol (MCP), RAG, PyTorch, LangChain, scikit-learn
Databases: PostgreSQL, MySQL, MongoDB, Qdrant